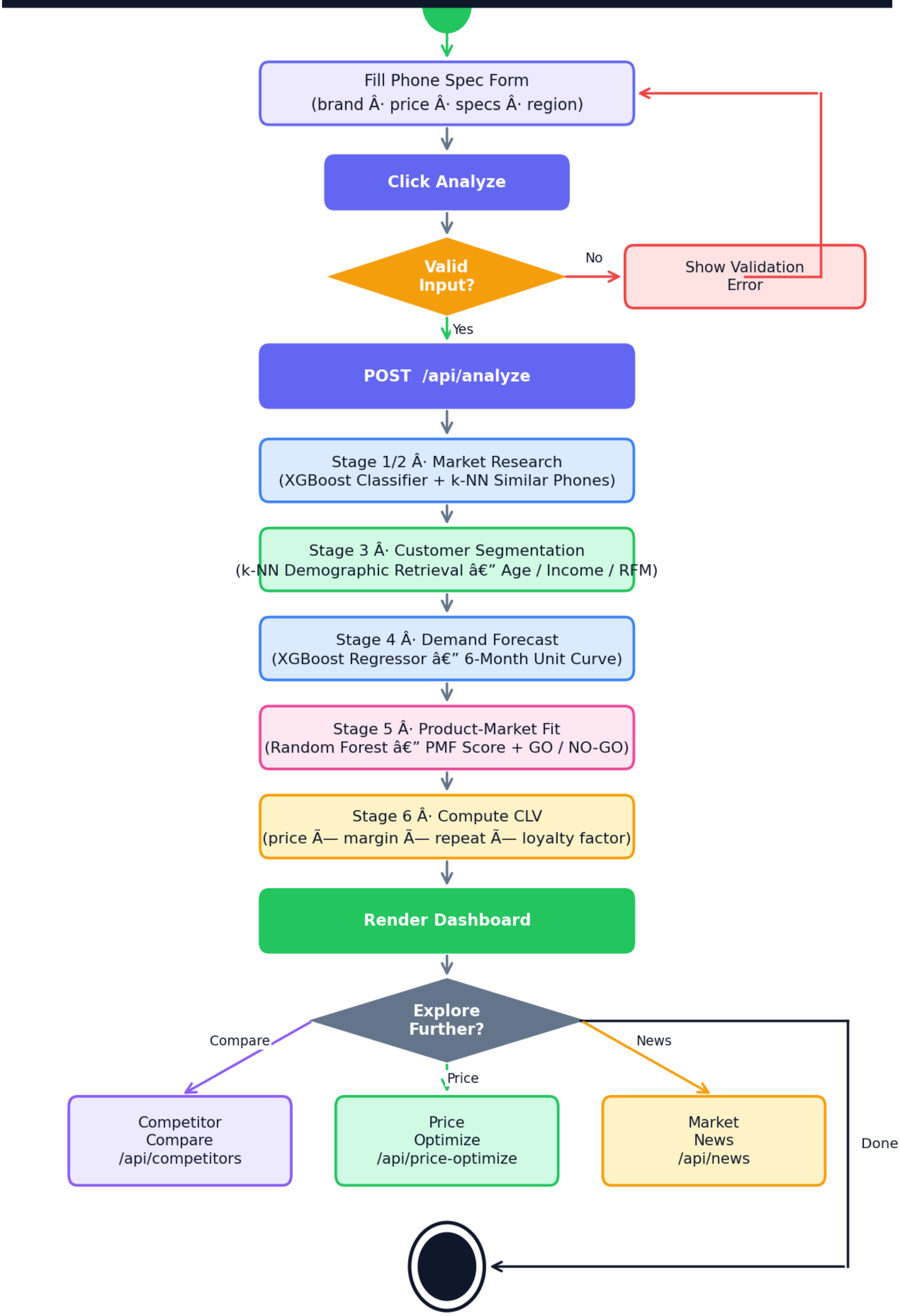
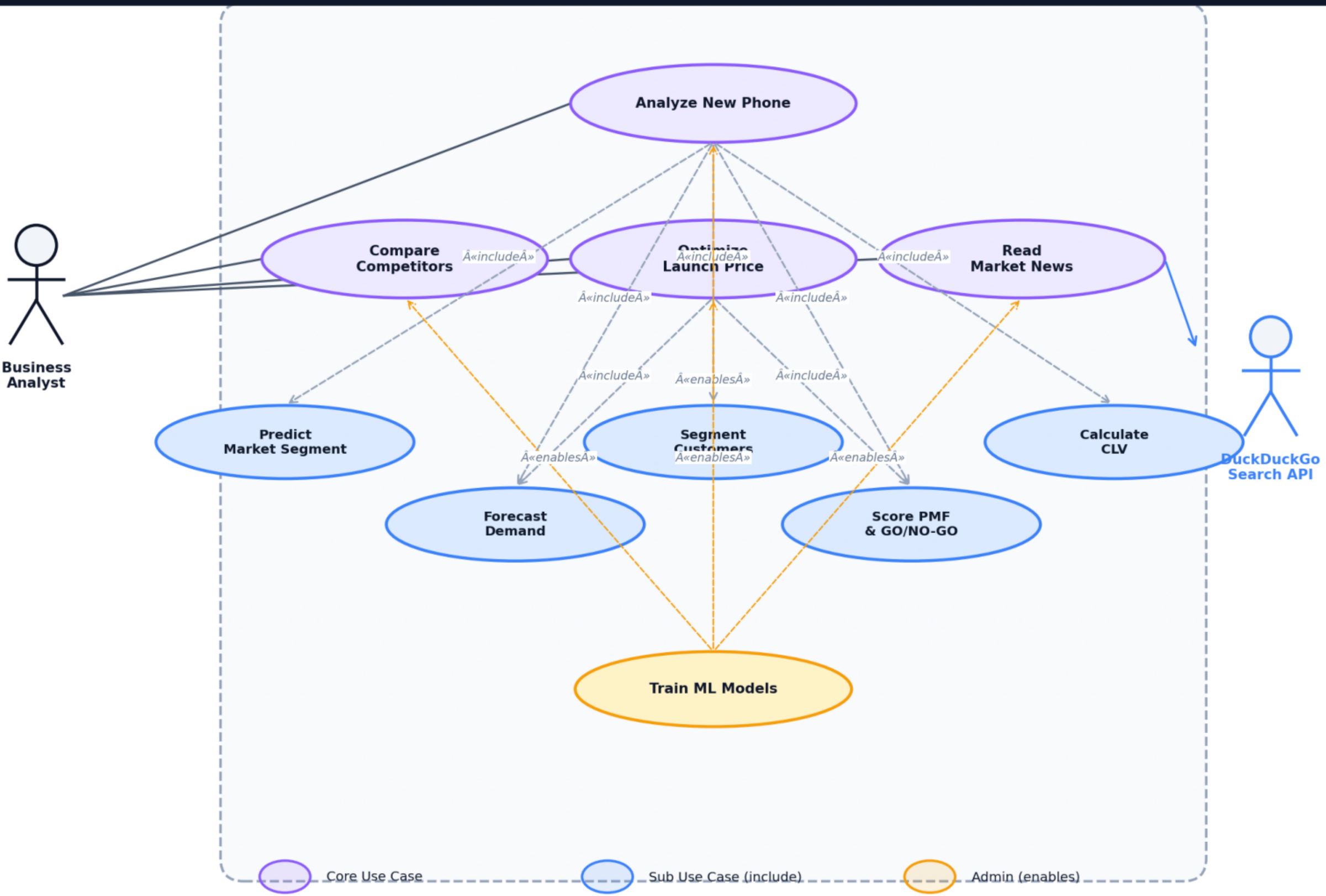
1. Activity Diagram

End-to-end user flow: spec input â†’ 6-stage BI pipeline â†’ dashboard



2. Use Case Diagram

Actors, use cases, and their relationships in the BI Agent system



Â«REST APIÂ»

3. Class Diagram

Modules, data models, and relationships across the BI Agent codebase

+analyze(spec, region)
+competitor_analysis(spec)
+price_optimize(spec)
+market_news(spec)
â€“spec_to_dict(spec)
â€“_ddg_search(q,n)

validates
calls
reads

Â«PydanticÂ»
PhoneSpec
+brand_name: str
+price_usd: float
+region: str
+five_g: int
+ram_capacity: int
+battery_capacity: int
+os: str

Â«OrchestratorÂ»
Agent
+analyze(spec, region, horizon)
+estimate_clv(price, curve, repeat)

Â«ModuleÂ»
Config
DATA_PATH: Path
MODEL_DIR: str
NUM_SPECS: List[str]
CAT_SPECS: List[str]
POST_LAUNCH: List[str]
+add_engineered_features(df)

uses
uses
uses
uses

Â«ML PillarÂ»
Pillar1 MarketResearch
Algorithm: XGBoost + k-NN
+train(df)
+predict_segment(spec)
+find_similar(spec, k)

Â«ML PillarÂ»
Pillar2 DemandForecast
Algorithm: XGBoost Regressor
+train(df)
+predict_curve(spec, region, horizon)

Â«ML PillarÂ»
Pillar3 Segmentation
Algorithm: k-NN k=10
+evaluate(df)
+predict_demographics(spec, region)

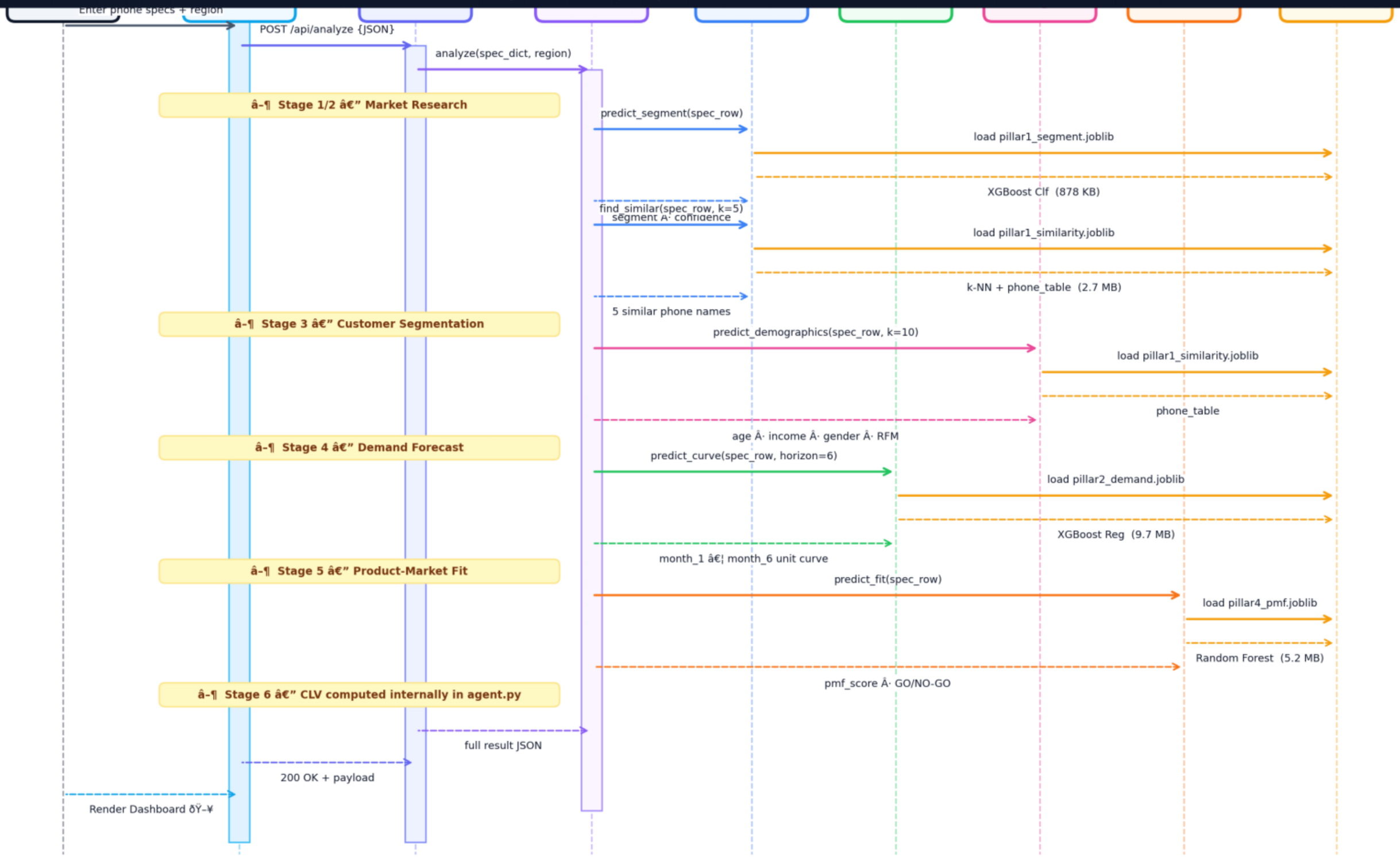
Â«ML PillarÂ»
Pillar4 PMF
Algorithm: Random Forest
+train(df)
+build_labels(df)
+predict_fit(spec)

Â«joblib FilesÂ»
ModelStore
pillar1_segment.joblib â€“ 878 KB (XGBoost Clf)
pillar1_similarity.joblib â€“ 2.7 MB (k-NN + table)
pillar2_demand.joblib â€“ 9.7 MB (XGBoost Reg)
pillar4_pmf.joblib â€“ 5.2 MB (RandomForest)

â”€â–¶ uses / calls
â”€â–¶ loads
--â–¶ direct API ref
â”€â–¶ reads config

4. Sequence Diagram

Message flow for POST /api/analyze â€” user submission to dashboard render



5. Data Flow Diagram

Data movement from raw CSV and user input through ML models to dashboard outputs

